



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

- The goal of this project was to predict the success of the SpaceX Falcon 9 first-stage landing, which is a key factor in reducing launch costs to approximately \$62 million per mission.
- Data collection was obtained through SpaceX REST API calls and Web Scraping of Wikipedia using BeautifulSoup.
- Dataset was cleaned and performed exploratory analysis using SQL to identify launch patterns.
- Folium maps was created to analyze site proximity to infrastructure and a Plotly Dash dashboard for real-time success rate filtering.
- Four classification models (Logistic Regression, SVM, KNN, and Decision Tree) were trained and tuned using GridSearchCV.
- All models achieved a consistent 83.3% accuracy on the test set.
- The confusion matrix revealed 0 False Negatives, meaning the model never failed to predict a successful landing when it actually occurred.

Introduction

- Commercial space age has arrived with emerging companies such as Virgin Galactic, Space X, and others.
- These companies are making space travel affordable for everyone...
- Other providers cost upwards of 165 million dollars each, SpaceX Falcon 9 rocket launch cost 62 million dollars.
- The savings is because SpaceX can reuse the first stage, but sometimes the first stage does not land.

Predicting whether the first stage of the rocket can be reused becomes a key factor to estimate the final price of each mission

Section 1

Methodology

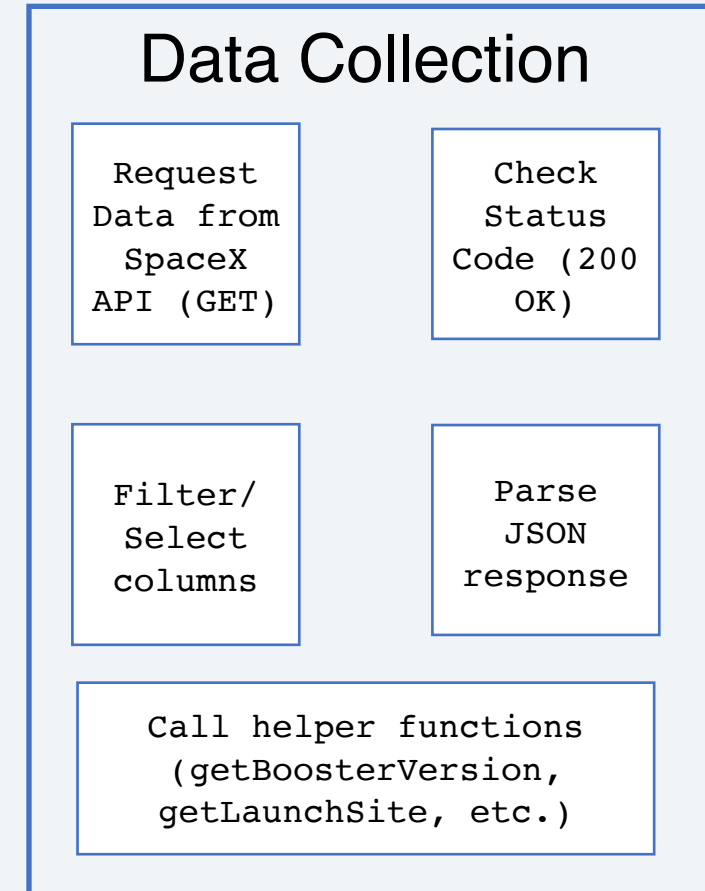
Methodology

Executive Summary

- Data collection methodology:
 - SpaceX Launch data was collected using GET request and web scraping (BeautifulSoup) from wikipedia.
- Perform data wrangling
 - Data was processed with pandas and numpy libraries in Python
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Classification models were tuned using GridsSearchCV and the accuracy of each model was obtained for model evaluation

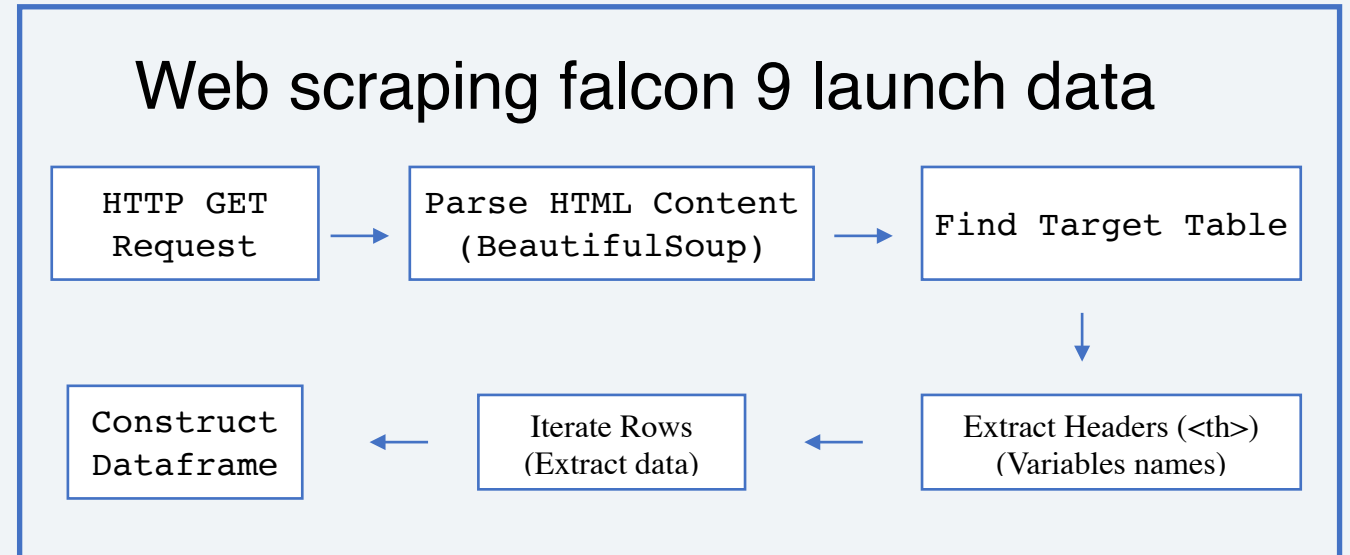
Data Collection – SpaceX API

- Data was obtained and parsed from SpaceX API using GET request
- The response was transformed to a data frame with `.Json_normalize`
- Launch information was obtained using the given ID's
- Some information was obtained with provided functions (`getBoosterVersion`, `getLaunchSite`, etc.)



Data Collection - Scrapping

- HTTP GET method was used to request falcon 9 launch page
- The request was assigned to a BeautifulSoup object
- Variables were extracted from the HTML table header
- A data frame was created by parsing the launch HTML tables



Data Wrangling

- Data frame was filtered to show falcon 9 data only.
- Missing values of PayloadMass variable were replaced with mean value.
- The categorical variable “Class” was created and added to the data frame with 0 and 1 as values indicating failure and success respectively.

EDA with Data Visualization

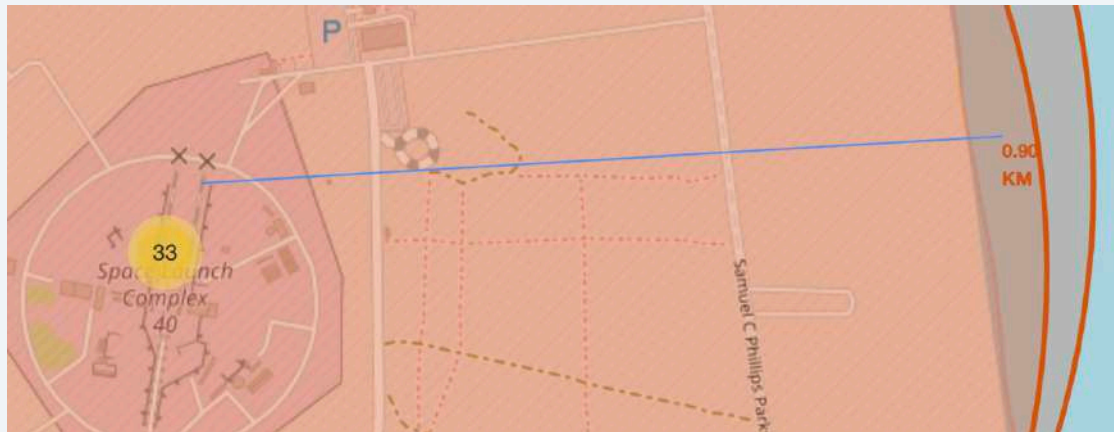
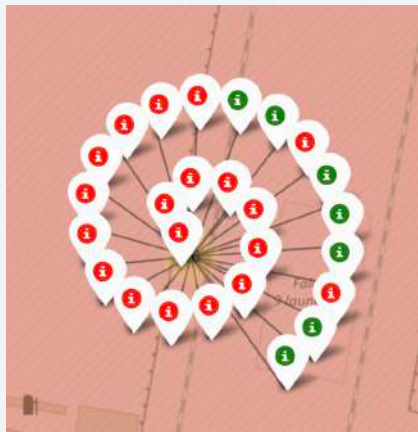
- Cat plots were used to visualize the relationship between discrete variables (like Flight Number) and the mission outcome (Class)
- Scatter plots were used to identify relationships between variables for example Payload mass vs Launch site or vs Orbit etc.
- Bar plot was used for visualizing success rate based on the type of orbit.
- Line plot was used for determining the change of success rate over the years.

EDA with SQL

- For EDA with SQL SpaceX data was loaded into a SQL table.
- Unique Launch sites were identified using DISTINCT query.
- The total Payload mass by boosters launched by NASA (CRS) was calculated using SUM query.
- Average Payload mass of booster version f9 v1.1 was obtained using AVG query
- Total number of failures and successes were obtained with COUNT query
- A list of booster versions with the maximum Payload mass was obtained using a subquery.
- Number of failure and success of specific periods of the year were obtained using subquery.

Build an Interactive Map with Folium

- Circle and markers were used to identify Launch sites activity, success and failure.
- PolyLines and distance markers were used to calculate proximity of Launch sites and critical infrastructure, like coastlines, highways, nearest cities etc.
- Color labels were used for better visualization of failure and success in every Launch site.

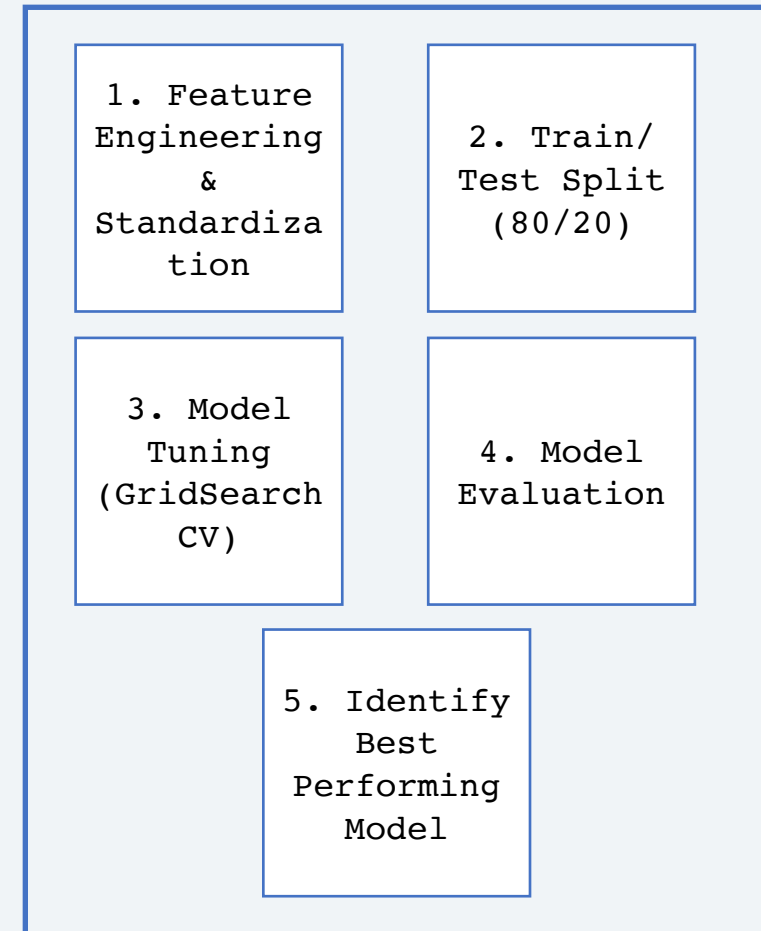


Build a Dashboard with Plotly Dash

- Pie charts were used to visualize total successful and failure launches for all sites and for a specific site.
- Scatter plots were created to show the relationship between Payload mass and launch success.
- Scatter plots were implemented with a slider to select a Payload mass range of interest.
- Callback functions were used to update the created plots for the selected input.

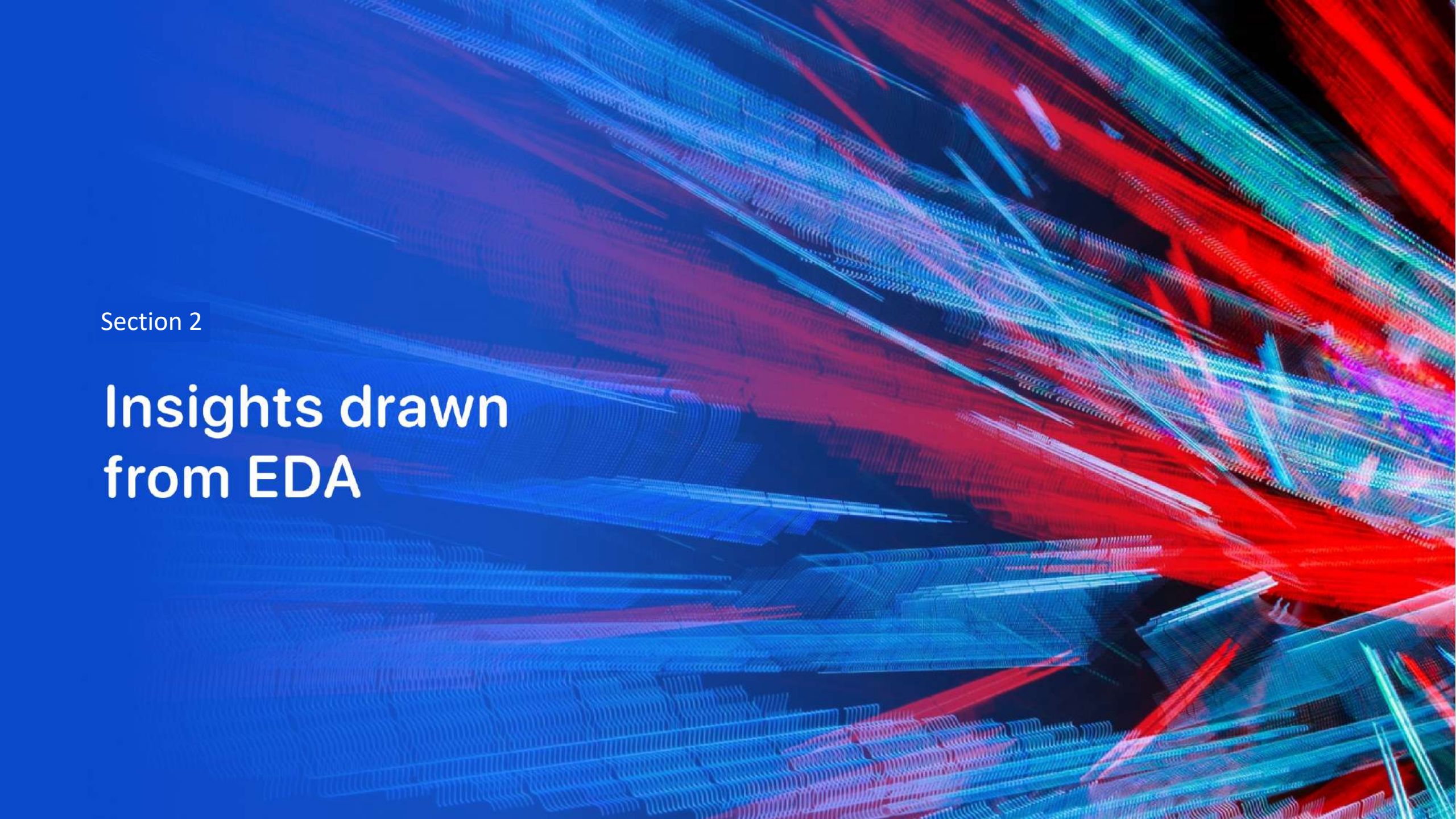
Predictive Analysis (Classification)

- Independent features and target variable (Class) were separated and assigned to X and Y objects respectively.
- Independent variables were standardized using StandardScaler to improve model performance.
- Data was split into training and testing with a test size of 0.2
- GridSearchCV was performed with multiple models to find their best parameters.
- Accuracy and confusion matrix of every model was obtained for model evaluation.



Results

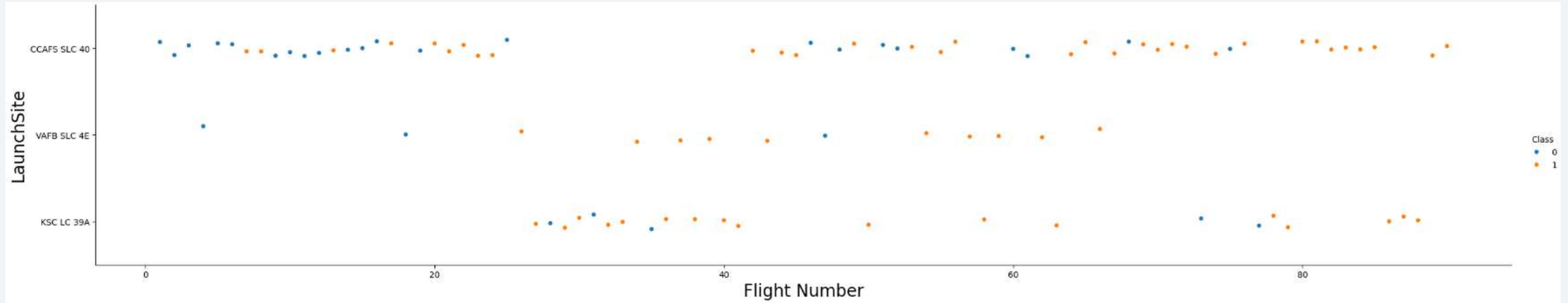
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



Section 2

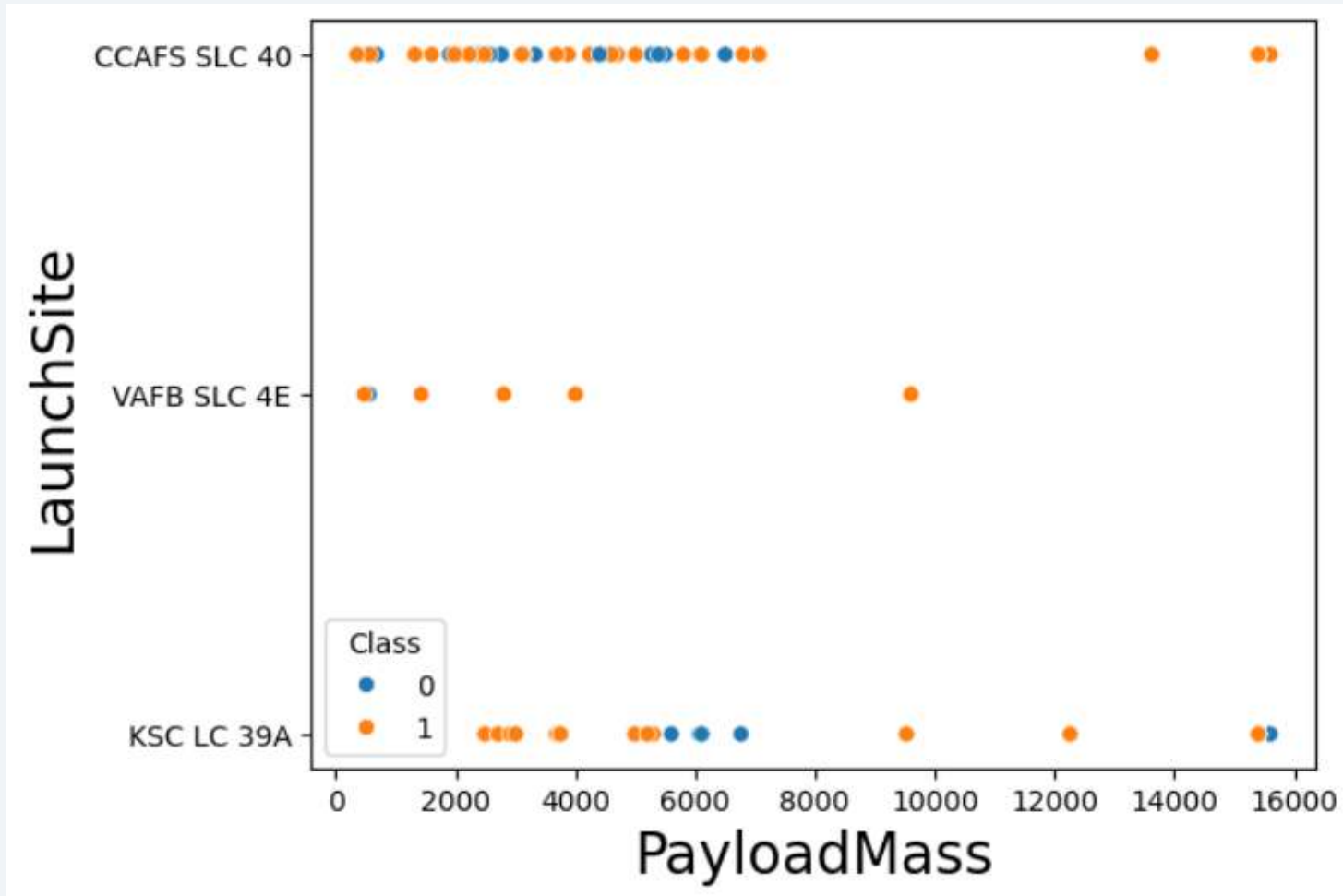
Insights drawn from EDA

Flight Number vs. Launch Site



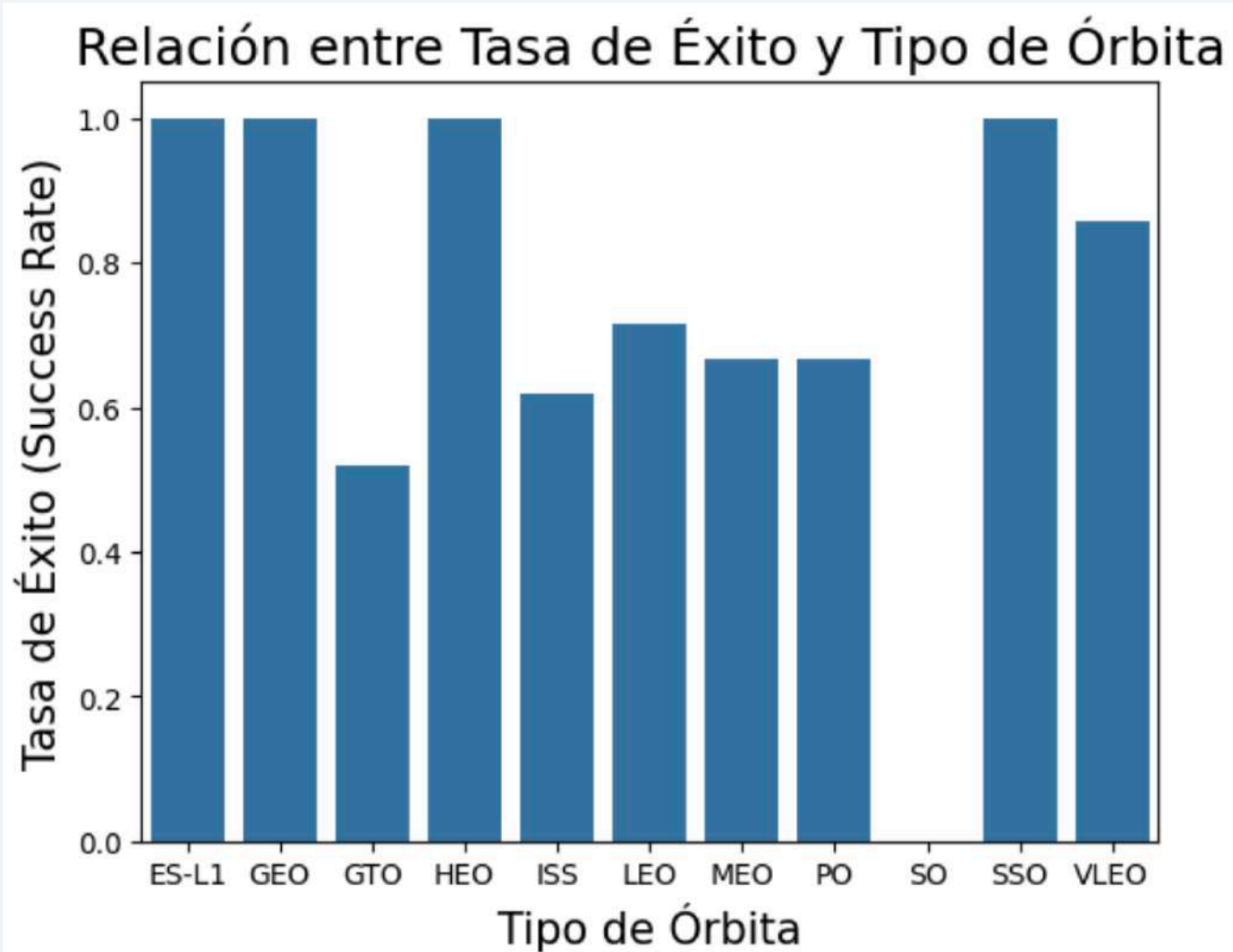
- From the plot above we can notice success of Falcon 9 launches increases over flight number.
- New Launch Sites were introduced or tested over flight number.

Payload vs. Launch Site



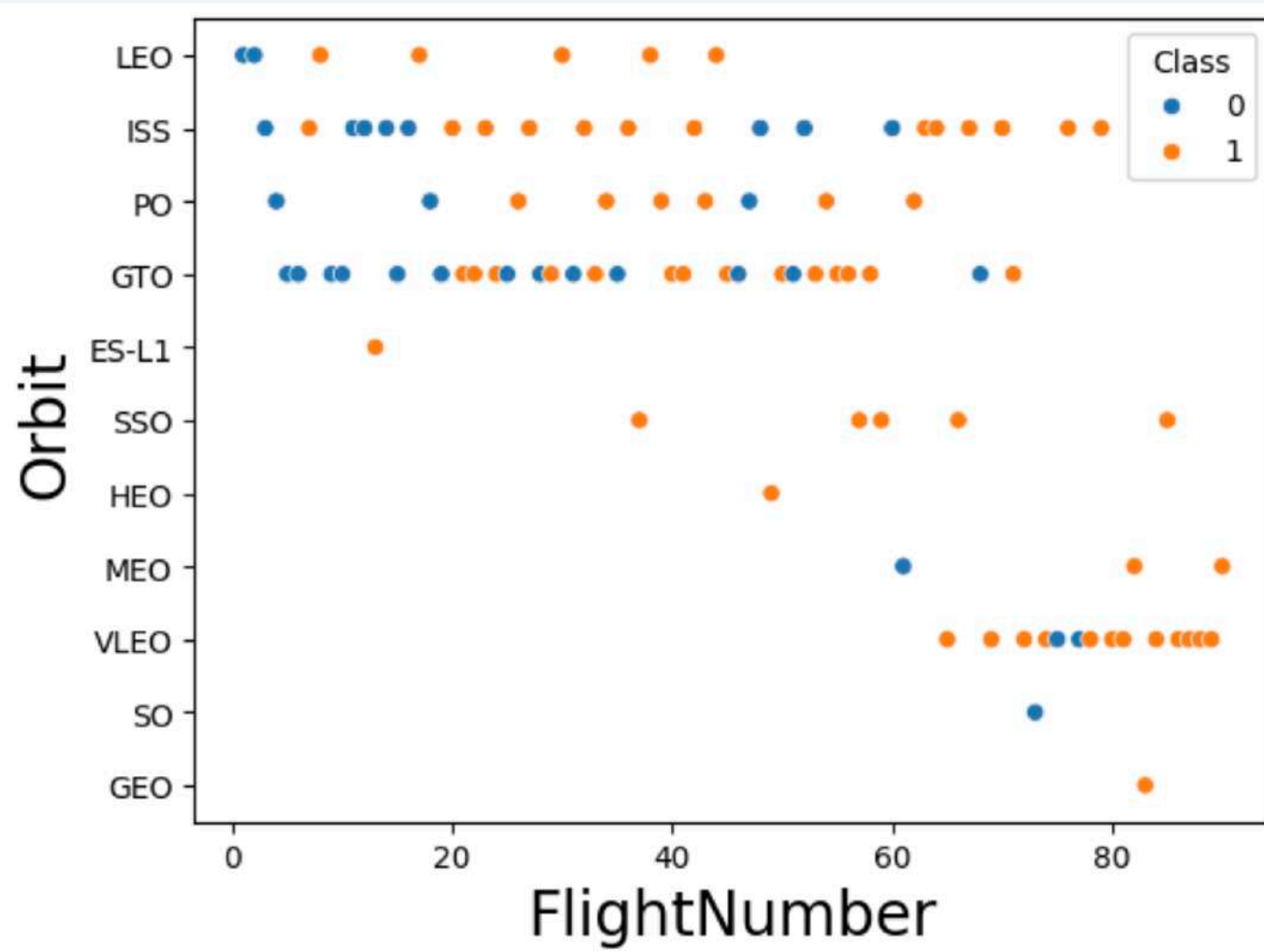
- VAFB SLC 4E Launch site hasn't been tested with payload mass Above 10000

Success Rate vs. Orbit Type



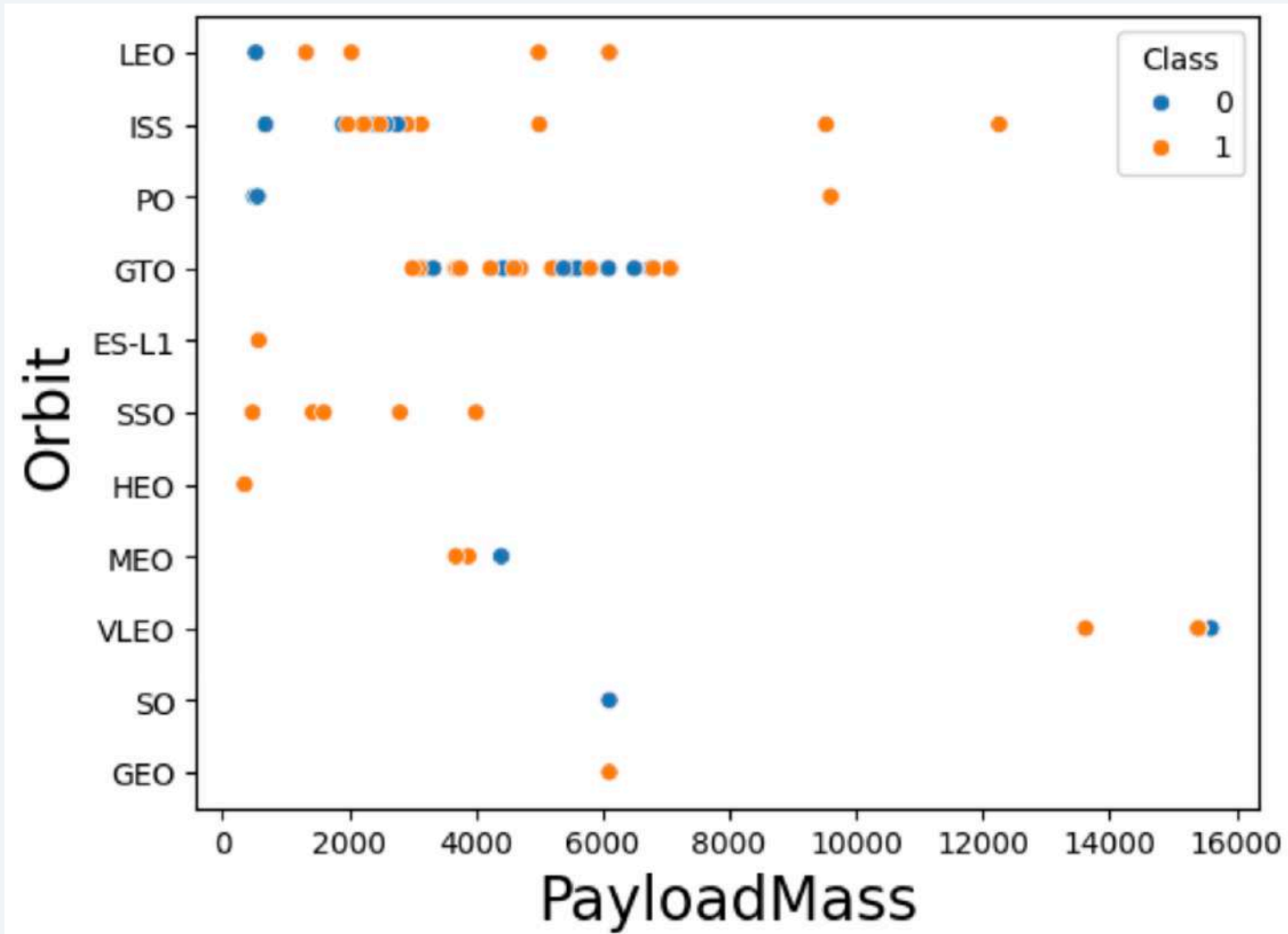
- ES-L1, GEO, HEO and SSO orbit types are the most successful.
- SO orbit has a 0 success rate.

Flight Number vs. Orbit Type



- GTO orbit doesn't show a clear pattern respect to flight number.
- LEO orbit becomes better over flight number
- SSO orbit hasn't had any failed launches, but more attempts would be suitable.

Payload vs. Orbit Type



- LEO, PO and ISS orbit types improve landings when increasing payload mass.
- GTO orbit doesn't show a clear pattern.

Launch Success Yearly Trend



- Success rate has been increasing since 2013.
- There was a notable decrease between 2017 and 2018.

All Launch Site Names

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

SpaceX has four Falcon 9 launch sites

Launch Site Names Begin with 'CCA'

Launch_Site

CCAFS LC-40

CCAFS LC-40

CCAFS LC-40

CCAFS LC-40

CCAFS LC-40

- Those are 5 launch sites beginning with “CCA”

Total Payload Mass

- Total payload mass carried by NASA (CRA) was 45596 Kg.

```
SUM(PAYLOAD_MASS__KG_)
45596
```

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1 was 2928.4 Kg.

```
AVG(PAYLOAD_MASS__KG_)
```

```
2928.4
```

First Successful Ground Landing Date

- First successful landing on a ground pad was on December 22 of 2015.

MIN(Date)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List of boosters that has successfully landed on drone ships and had a payload mass greater than 4000 but less than 6000.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- It was found 99 successful mission outcomes, 1 in failure and 1 with payload status unclear.

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Boosters that has carried the maximum payload mass are shown below.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- Failed landing outcomes in year 2015.
- Both landings were on drone ship.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Drone ship has more attempts with successful (5) landings but also with failures (5).
- Ground pad has successful landings only.

Landing_Outcome	Count_Launch
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

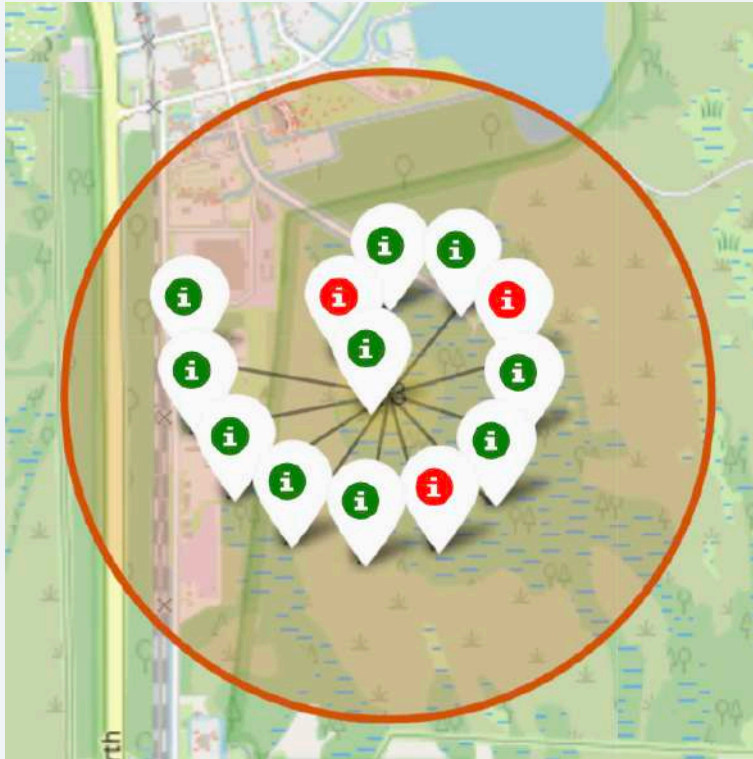


Launch sites location map



- From the map we can notice all launch sites are near the coastlines.
- Launch sites in Florida are nearest to the equator.

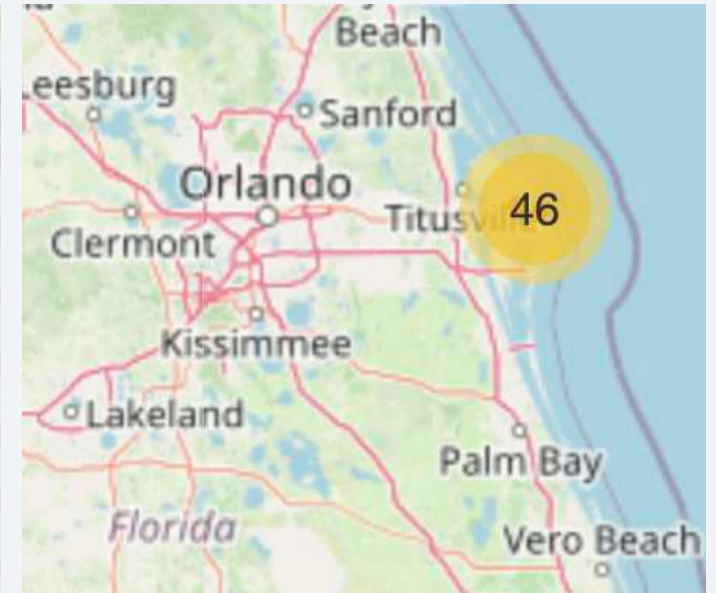
Launch outcomes



- KSC LC - 39A has more successful landings



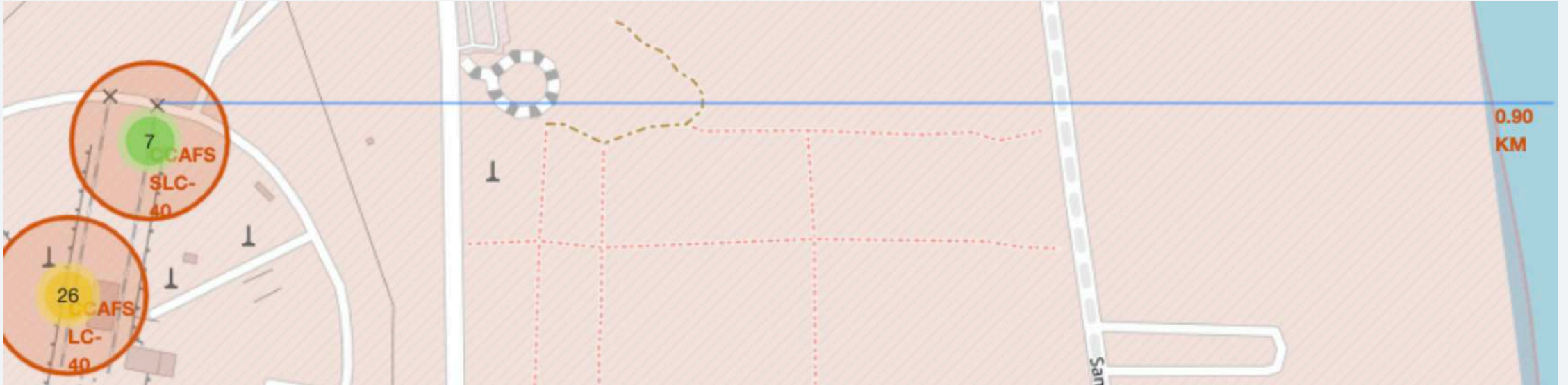
California (10)



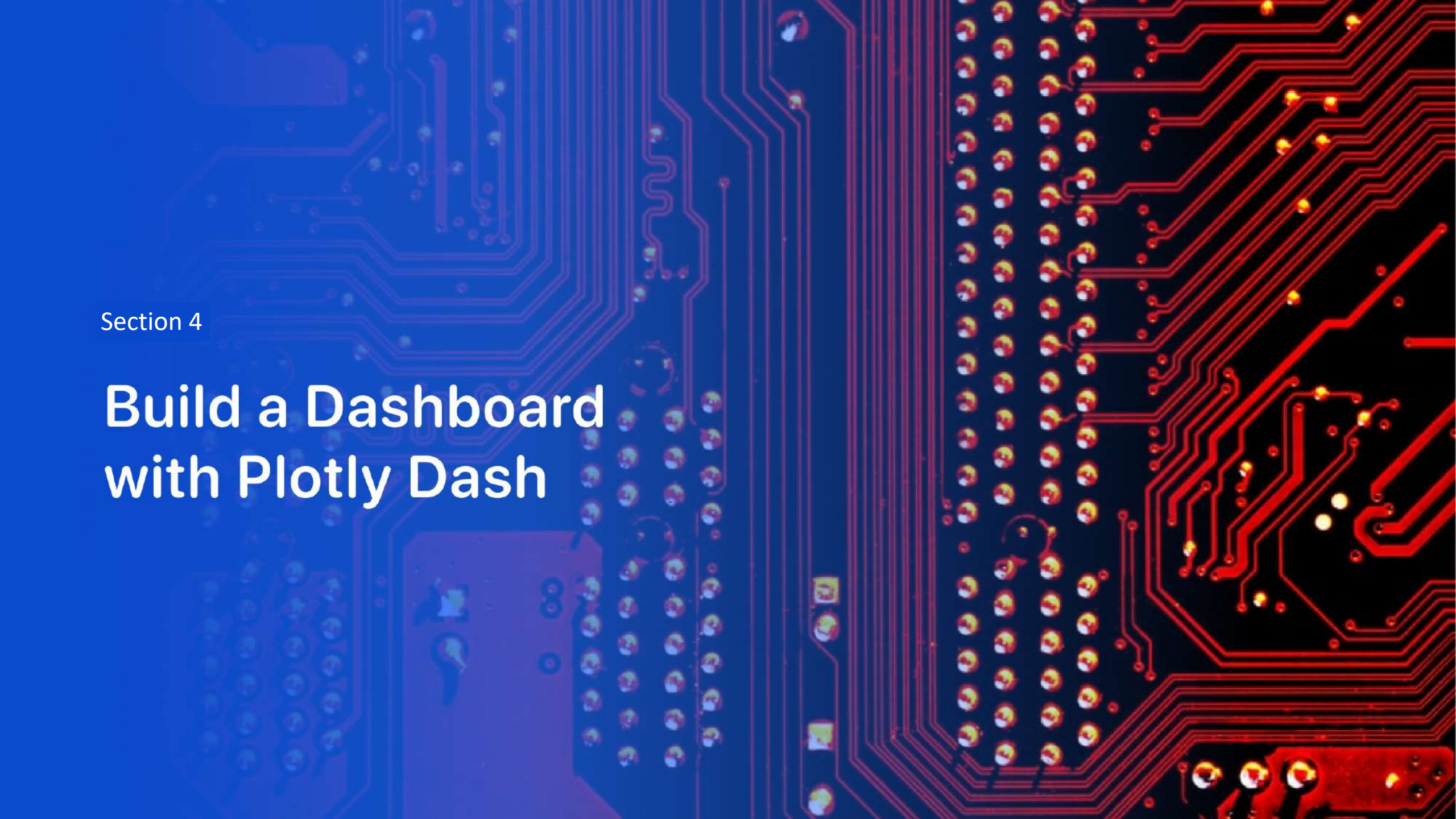
Florida (46)

- California has less flights and launch sites than Florida

Launch sites proximity area



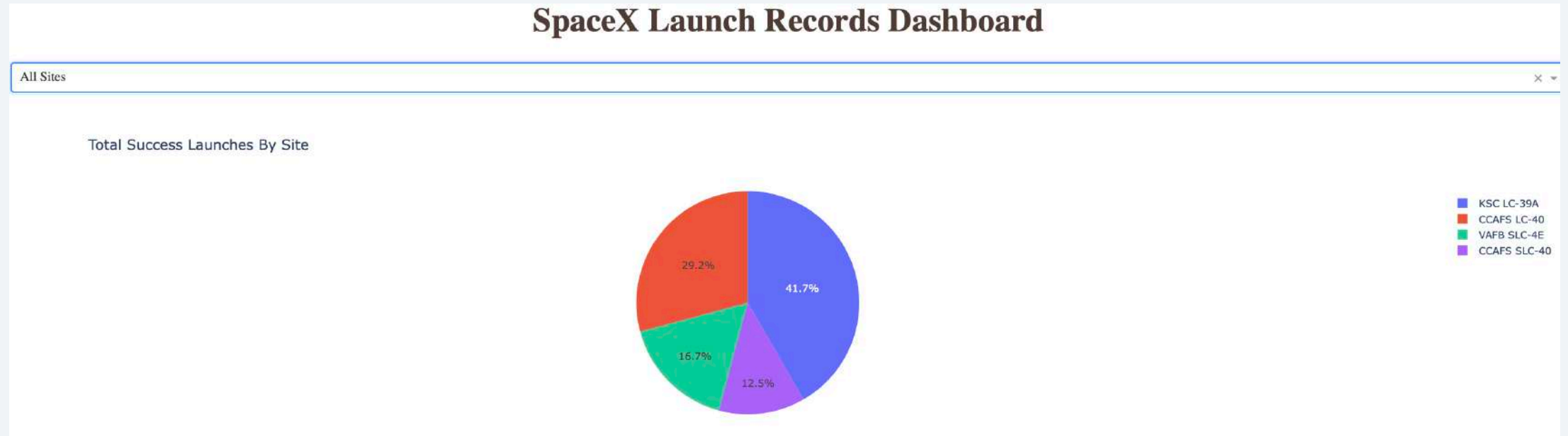
- Launch sites are near coastlines and far from cities.
- Proximity to the nearest coastline in Florida is 0.9Km approximately.
- Launch sites are near highways for transportation logistics.



Section 4

Build a Dashboard with Plotly Dash

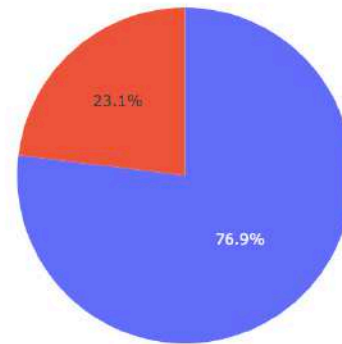
Total success launches by site



- KSC LC-39A has the most successful launches with 41.7%
- CCAFS SLC-40 has the least successful launches with 12.5%.

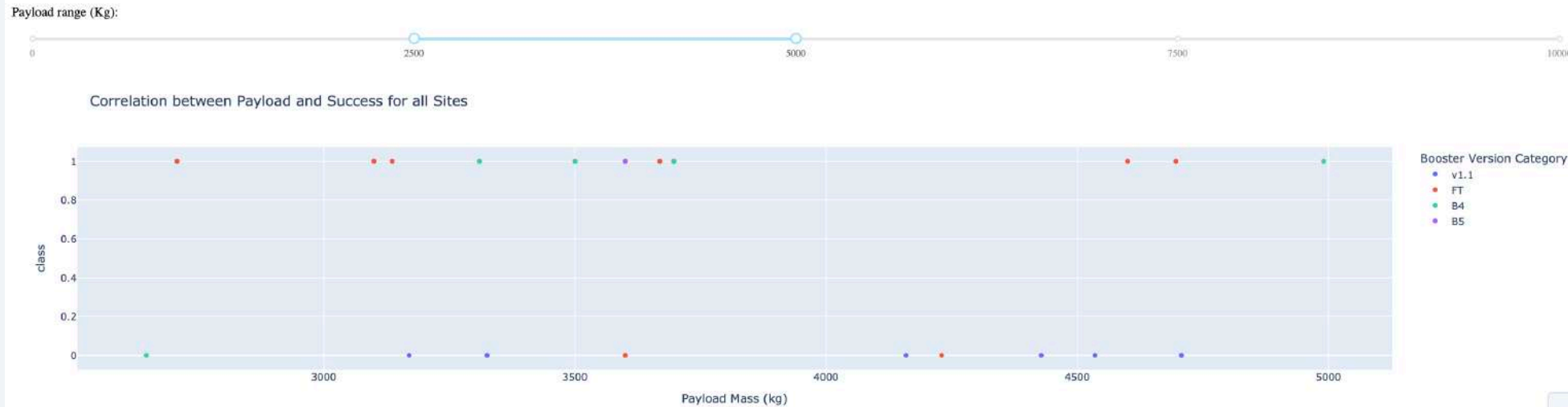
KSC LC-39A Launch site

Total Success Launches for site KSC LC-39A

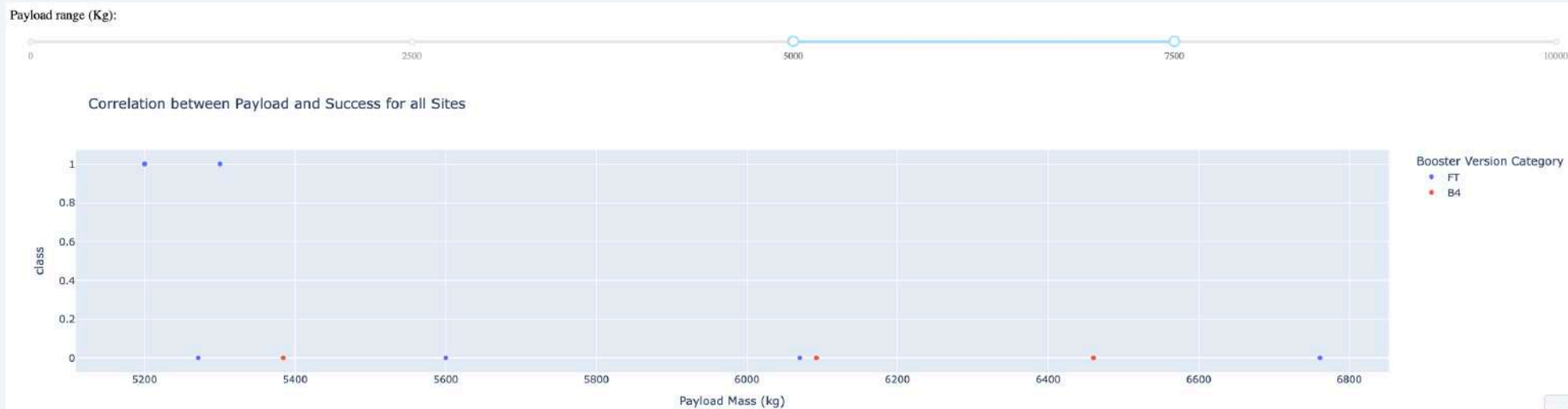


- From the most successful launch site, 23.1% had a failed outcome.

Payload mass vs success rate for all sites



- 2500 to 5000 payload mass (Kg) range has the most successful outcomes.



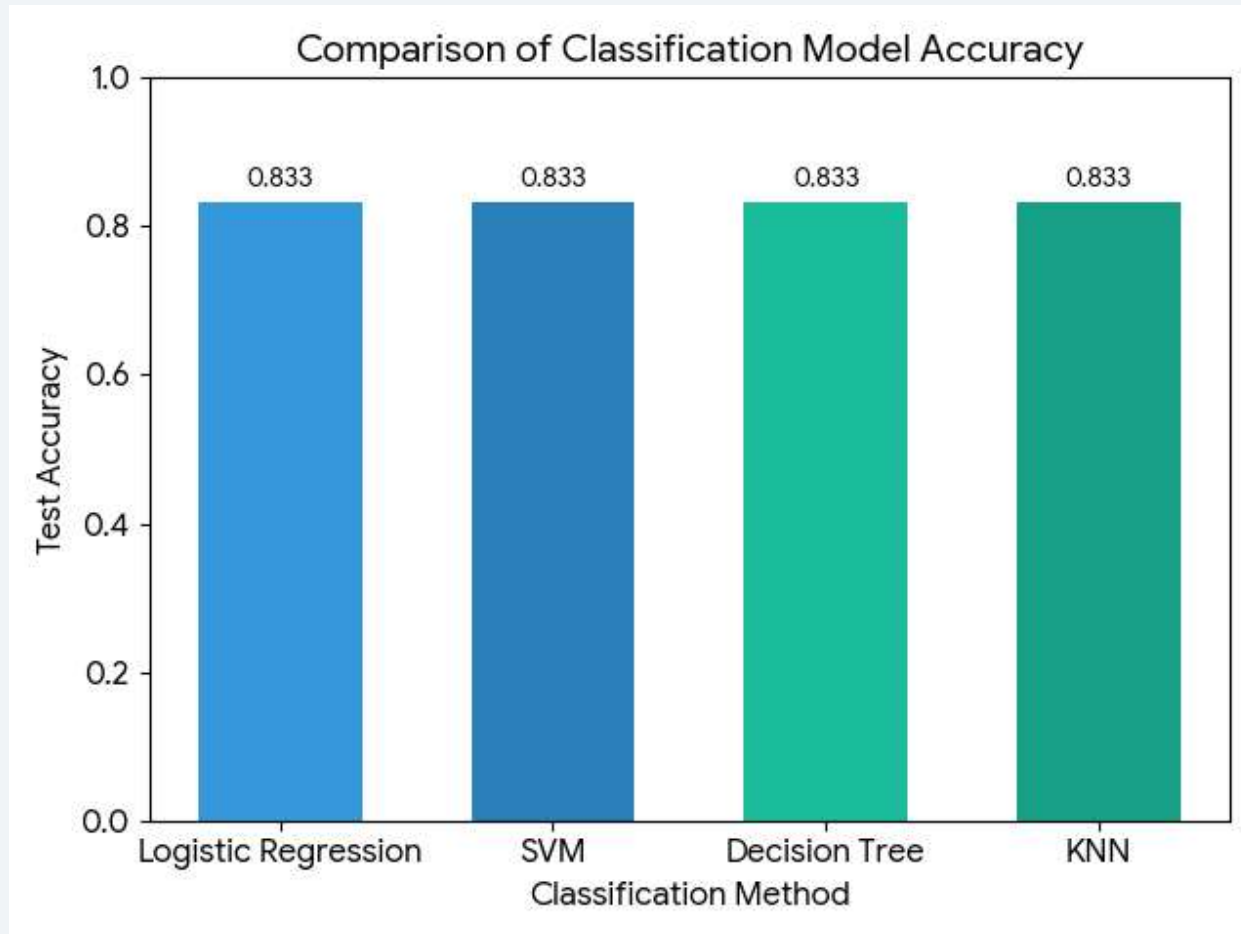
- 5000 to 7000 payload mass (Kg) range has the least successful outcomes.



Section 5

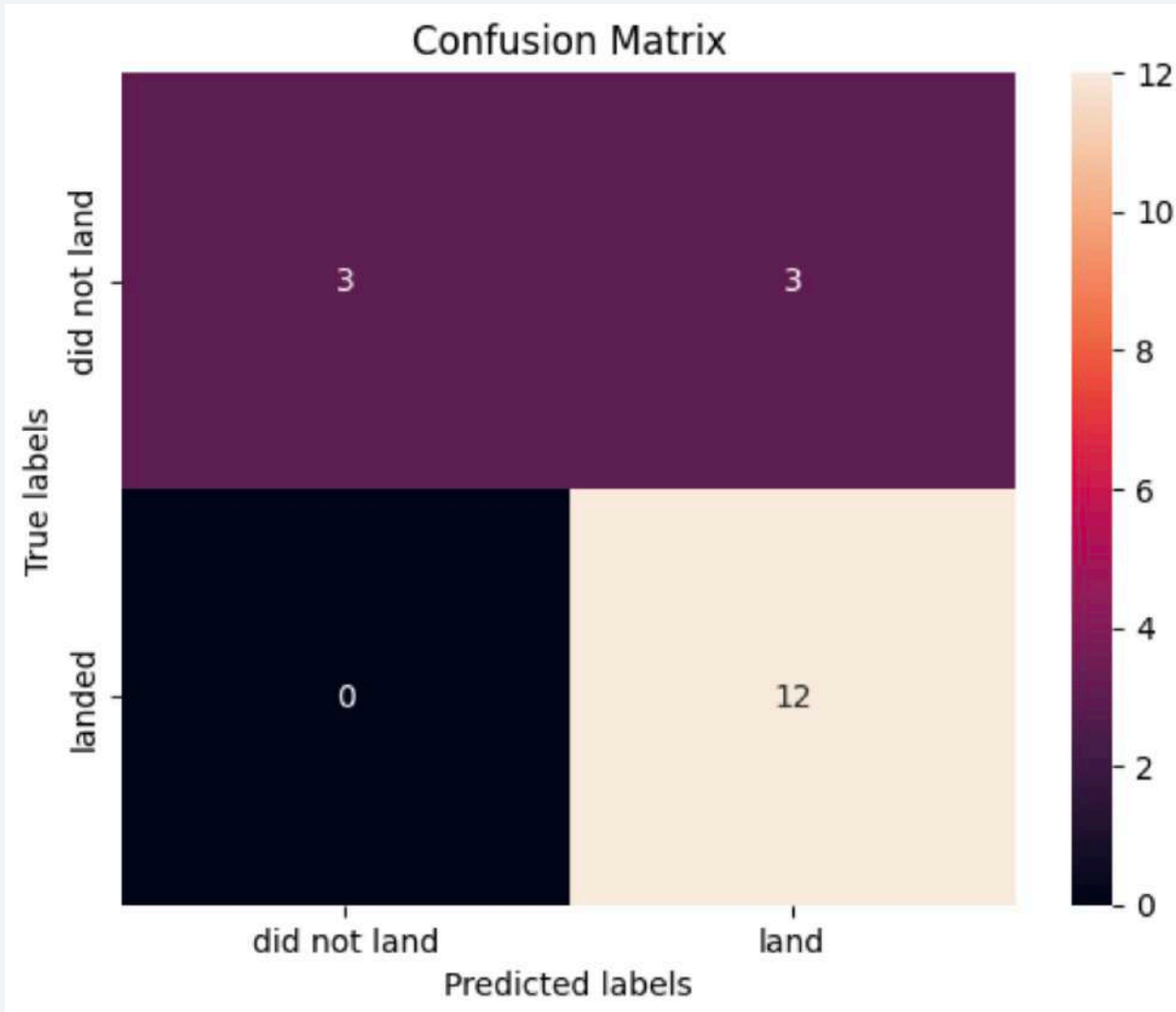
Predictive Analysis (Classification)

Classification Accuracy



- All tested models resulted in the same accuracy score (0.833)

Confusion Matrix



- 0 False negatives (landed that didn't land)
- 3 True negatives (Actually didn't landed)
- 3 False positives (didn't landed that landed)
- 12 True positives (Actually landed)

Conclusions

- Exploratory analysis confirms a significant positive correlation between flight number and success rate.
- Geospatial analysis via Folium proved that all launch sites are strategically located.
- All machine learning models (Logistic Regression, SVM, KNN, and Decision Tree) achieved an identical accuracy of 83.3%
- The Confusion Matrix demonstrates that the model is particularly effective at identifying successful landings (0 False Negatives)

Appendix

```
df.head(5)
```

Python

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

```
import pandas as pd
df = pd.read_csv("https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/labs/module_2/data/Spacex.csv")
df.to_sql("SPACEXTBL", con, if_exists='replace', index=False, method="multi")
```

101

Note: This below code is added to remove blank rows from table

```
#DROP THE TABLE IF EXISTS

%sql DROP TABLE IF EXISTS SPACEXTABLE;

* sqlite:///my_data1.db
Done.

[]

%sql create table SPACEXTABLE as select * from SPACEXTBL where Date is not null

* sqlite:///my_data1.db
Done.
```

```
parameters = {'C': [0.01, 0.1, 1],
              'penalty': ['l2'],
              'solver': ['lbfgs']}
```

```
parameters = {'C': [0.01, 0.1, 1], 'penalty': ['l2'], 'solver': ['lbfgs']} # l1 lasso l2 ridge
lr = LogisticRegression()

logreg_cv = GridSearchCV(lr, parameters, cv=10)
logreg_cv.fit(X_train, Y_train)
```

GridSearchCV

best_estimator_:
LogisticRegression

LogisticRegression

Thank you!

